

ERIKA BUSSMANN

Engineer / Researcher

Mountain View, CA | erikabusmann@google.com

ENGINEERING & RESEARCH EXPERIENCE

Research Engineer

March 2026 – Present

Human-AI Interaction, GDM Responsibility

- Design and iterate on auto-evals for detecting interaction dynamics in human-model conversations, calibrating against human-labeled data to build golden datasets
- Perform in-depth qualitative review of user-model logs to inform evaluation metrics
- Apply time-series statistical analyses to conversational-geometry features (e.g. TRACE, embedding-trajectory metrics) to model the presence and development of specific interaction dynamics across user-model conversations
- Extend this methodology to additional interaction dynamics relevant to potential harms, with results in preparation for publication

Software Engineer

July 2025 – Present

Gemini Live in Chrome, GDM GeminiApp

- Lead Gemini Live model release evaluation efforts for the Chrome surface
- Design primary eval sets and metrics for model quality hill-climbing and regression detection
- Lead experiments to identify optimal orchestration of sub-agents (e.g. for tool selection)
- Design a framework to root-cause and address tool call over-triggering via CoTs, reducing over-triggering, compute costs, and latency
- Design, run, and land experiments optimizing browser context truncation strategies
- Design and implement an automated process to enable regular refresh of critical benchmarks from user logs, now a key part of the evaluation and improvement pipeline

RESPONSIBLE COMPUTING & AI SAFETY ENGAGEMENT

Co-Instructor

May 2025 – Present

“Capital” For Tech Workers

- Co-designed/taught a course to help tech workers critically examine the ethics and political economy of the tech industry, with AI as a central case study
- Co-authored the corresponding textbook in partnership with researchers at the Free Computing Lab

AI Principles Advisor

May 2023 – May 2025

Google AI Principles Pioneers

- Trained contributor to adversarial testing of new launches, evaluating fairness, safety, and human-rights-related risk vectors against Google’s AI Principles

Responsible Computing Instructor

July 2021 – May 2022

Deep Learning — Brown University

- Developed curriculum to help future tech workers critically engage with the sociopolitical, economic, and environmental consequences of deep learning-powered products
- Led weekly labs and conceptual/debugging office hours for 200+ students
- Invited to guest lecture on the practice of responsible computing to 250+ students

Co-Creator & Instructor

July 2020 – Dec. 2020

Technofutures & Dystopias — Brown University

- Designed and led an independent survey study on designing tech futures, exploring automation and the future of work, AGI, and tech workers’ rights

ADDITIONAL ENGINEERING EXPERIENCE

Software Engineer

July 2024 – Sept. 2025

Auction Optimization, Google Ad Exchange (AdX)

- Designed and launched real-time bidding throttling optimizations, including sizing opportunity headroom via simulation and leading live traffic A/B experiments
- Worked cross-functionally with research engineers, PMs, legal counsel, and bidders

Software Engineer

July 2022 – Sept. 2024

Efficiency & Obtainability, Google Compute Engine (GCE)

- Designed, executed, and owned the primary data layer for Project Moose; named inventor on Patent GP-307734-00-US, “Cloud Obtainability Optimization & Stockout Experience”
- Independently designed and launched a source of truth for a critical zone-isolation reliability initiative, relied on by major GCP products for risk compliance

EARLY RESEARCH EXPERIENCE

Research Engineer

2017 – 2019

UMN CS & Engineering, Medical Computer Vision Lab

- Designed and developed melanoma detection classifiers via novel morphological features
- Collaborated with dermatologists and imaging experts to understand problem space
- Contributed original research to the International Skin Imaging Collaboration Benchmarks

Applied Math Researcher

2016 – 2018

UMN Department of Mathematics

- Conducted original research in applied dynamical systems under Ph.D. candidate Kate Meyer (Mathematics) and Prof. Clarence Lehman (Computational Biology)
- Published co-author, “Resilience of socially valued properties of natural systems to repeated disturbance,” in *Nature (Sustainability)* and *Natural Resource Modeling*

EDUCATION

B.S., Computer Science & Applied Mathematics

Brown University, Providence, RI

SKILLS & RESEARCH INTERESTS

Languages	C++, Python, SQL, MATLAB
Tools / Libraries	TensorFlow, Pandas, NumPy, Scikit-learn
Research Interests	Evaluation Science, Scalable Oversight, Socioaffective AI Safety